

CYBEX – CYBERCRIME PREDICTIVE INTELLIGENCE

DATABASE ARCHITECTURE, DATA SCHEMAS & VIVA PRESENTATION GUIDE

Project: CybEx Cybercrime Intelligence	Database Engine: SQLite 3 (Relational RDBMS)	Total Active Records: 62,700+
ORM Layer: Sequelize ORM (Node.js)	ML Microservice: Python (Scikit-Learn)	Date: August 2026

1. Teacher / Examiner Ko Kya Batana Hai? (Quick Viva Answer)

■ **30-Second Perfect Answer for Teacher:**
"Sir/Ma'am, humne is project me **Relational Database Management System (RDBMS)** ke roop me **SQLite** use kiya hai, jisko **Sequelize ORM (Object-Relational Mapping)** ke through Node.js Express backend se seamlessly connect kiya gaya hai. Database me total **10 structured tables** hain jinme **62,700+ records** hain (including 55,254 nationwide cybercrime complaints, 4,249 ATMs, real-time alerts, tasks, audit logs, and encrypted evidence documents). Iske sath Python Machine Learning model ko real-time risk prediction aur ML evaluation ke liye SQLite database ke sath integrated rakha gaya hai as a single source of truth."

2. Backend Database Inventory (Tables & Live Record Counts)

Humare database (server/database.sqlite) me total 10 tables hain:

Table Name	Record Count	Primary Key	Functional Description in Project
Complaints	55,254	id (Auto) / complaintId	Central registry of all verified cybercrime FIRs across 28 states & 8 UTs (UPI, Phishing, ATM Skimming, etc.).
ATMs	4,249	id / atmId	Geocoded ATM kiosks, dispensers, CCTV status, nearby vulnerability scores for GIS heatmap.
Alerts	201 (127 Active)	id	Real-time threat alerts with risk levels (CRITICAL, HIGH, MEDIUM), time window & target location.
Tasks	1,000	id	Officer workflow assignments (Freeze Mule Account, Verify KYC, Contact Victim, etc.).
AuditLogs	1,000	id	Tamper-evident legal audit log capturing officer logins, case edits, report exports.
Notifications	500	id	Role-based notifications for Officers, Citizen Portal, and Bank Nodal Officers.
Messages	500	id	Inter-officer & inter-bank secure communication regarding active mule chains.
Reports	Persistent (Dynamic)	id	Generated officer intelligence dossiers, crime summaries, state/district custom reports.

Documents	Persistent (Vault)	id	Secure evidence vault storing citizen transaction slips, audio/video evidence, KYC docs.
Users & PassReset	Active Accounts	id / userId	Bcrypt-hashed authentication credentials, role RBAC (Officer, Bank, Citizen), and OTP reset tokens.

3. Detailed Schema Specifications (Core Tables)

Table 1: Complaints (Core Dataset)

Field Name	Data Type	Constraints	Example / Description
id	INTEGER	PRIMARY KEY, AUTOINCREMENT	1, 2, 3...
complaintId	VARCHAR(255)	UNIQUE, INDEXED	CC-2026-0001, CC-2026-0042, CC001
type	VARCHAR(255)	NOT NULL	UPI Fraud, Phishing & SIM Swap, ATM Skimming
amount	VARCHAR(255)	NOT NULL	₹85,000, ₹2,50,000
location	VARCHAR(255)	NULLABLE	Andheri East, Mumbai, Maharashtra
state	VARCHAR(255)	INDEXED	Maharashtra, Uttar Pradesh, Delhi
district	VARCHAR(255)	INDEXED	Mumbai, Pune, Lucknow, Bengaluru
latitude / longitude	FLOAT / REAL	NULLABLE	19.1136, 72.8697 (GIS Coordinates)
status	ENUM / VARCHAR	DEFAULT 'Pending'	Pending, Analyzed, Under Investigation, Resolved
victimBank	VARCHAR(255)	NULLABLE	State Bank of India, HDFC Bank, ICICI Bank
suspectMule	VARCHAR(255)	NULLABLE	AC-78945612 (Mule Cluster #4)
predictionData	JSON / TEXT	NULLABLE	{ score: 87, riskLevel: 'HIGH', confidence: '94%' }
date / time	VARCHAR(255)	NULLABLE	28 Aug 2026, 18:45
createdAt / updatedAt	DATETIME	NOT NULL	Automatic timestamp tracking

4. Architecture & Single Source of Truth Pipeline

Is system ka sabse bada technical strength yeh hai ki har ek interface (Officer Dashboard, Citizen Portal, Bank Nodal Desk, GIS Heatmap, ML Prediction) alag-alag dummy arrays use nahi karta, balki **ek centralized backend database** se sync rehta hai:

Layer	Technology	Responsibility & Role
Frontend UI	React 18 + Vite + Leaflet GIS	Server-side pagination (50 items/page), deep-linking URL parameters, real-time map radar rendering.
REST API Server	Node.js + Express (Port 3001)	Exposes /api/cases, /api/alerts, /api/dashboard/stats, /api/ml/evaluation, JWT auth sessions.
ORM Layer	Sequelize ORM	Type safety, automated schema migrations, SQL query building, connection pooling.
Database Engine	SQLite 3 (database.sqlite)	Persistent ACID-compliant relational storage storing 62,700+ records with zero external database daemon overhead.
ML Microservice	Python Flask + Scikit-Learn (Port 5000)	Gradient Boosting Classifier trained on 55,254 complaints; generates real evaluated 99.98% accuracy on 11,051 test samples.

5. Top 10 Viva / Teacher Presentation Questions & Answers

Q1: Which database have you used in this project and why?

Answer: Humne Relational Database Management System (RDBMS) ke roop me SQLite 3 use kiya hai using Sequelize ORM. SQLite ko select karne ka reason yeh hai ki yeh serverless, self-contained, ACID-compliant, aur high performance hai. Isme bina kisi heavy database installation ke poora dataset (62,700+ records) lightweight file ke roop me embedded rehta hai, jo enterprise law enforcement portal ke local deployment ke liye ideal hai.

Q2: How does the system handle 55,000+ complaints without slowing down the React UI?

Answer: Humne Server-Side Pagination aur Search Indexing implement kiya hai. Frontend ek sath poore 55k records ko browser memory me load nahi karta, balki GET /api/cases?page=1&limit=50 ke through chunk-by-chunk 50 records fetch karta hai. Is wajah se frontend lightning-fast render hota hai aur memory utilization minimum rehta hai.

Q3: What is the relationship between Complaint, Prediction, Alert, and Map?

Answer: Complaint ID is the single anchor thread. Jab ek complaint (jaise CC-2026-0001) par Python ML prediction chalta hai, toh uski risk score, level aur geographic coordinates (Latitude, Longitude) calculate hoti hain. Agar risk score $\geq 70\%$ hota hai, toh backend automatically Alerts table me live Alert insert karta hai. Jab officer 'View on Map' click karta hai, toh URL me exact coordinates deep-link ho kar Heatmap us location par zoom hota hai.

Q4: What machine learning model is trained and what is its accuracy?

Answer: Humne Scikit-Learn ka GradientBoostingClassifier train kiya hai on 55,254 complaints using an 80/20 train/test split. 11,051 unseen test samples par evaluate karne par model ne 99.98% Accuracy, 0.9998 Precision, 0.9998 Recall, aur 0.9998 F1-Score deliver kiya hai, jo GET /api/ml/evaluation API par expose hota hai.

Q5: How are user passwords and authentication tokens stored in the database?

Answer: Passwords database me plain text me store nahi hote. Humne bcrypt hashing (salt rounds: 10) use kiya hai. Login ke waqt password hash match hone par backend JWT (JSON Web Token) generate karta hai jo frontend ke session state aur role-based access control (Officer, Bank, Citizen) ko secure karta hai.

Q6: Can SQLite be replaced with PostgreSQL or MySQL in production?

Answer: Yes, absolutely. Kyunki humne Sequelize ORM use kiya hai, hume sirf config connection string change karni hogi (e.g. postgres://user:pass@host/db). Database schema aur saari queries automatically PostgreSQL ya MySQL ke sath compatible hain bina kisi code rewrite ke.

Q7: Where are files and evidence stored?

Answer: Evidence files (PDFs, images, transaction slips) disk storage par /uploads directory me securely store hoti hain, aur unke cryptographic metadata (filename, size, mimetype, uploadedBy, linkedCaseId, timestamp) database ke Documents table me store hote hain.

Q8: How does the Officer Reports module work with the database?

Answer: Jab officer kisi State ya District ka custom report generate karta hai, backend dynamically Complaints database se aggregated metrics (total loss, top crime categories, status distribution) calculate karke Reports table me permanently store karta hai.

Q9: What is the difference between SQLite and NoSQL (MongoDB) for this use case?

Answer: Cybercrime financial records, FIR numbers, bank mule accounts aur legal audit trails me strict relational integrity aur ACID compliance mandatory hoti hai. SQLite structured tabular schema and foreign relations ensure karta hai.

Q10: What are the main tables in your system?

Answer: Humare system me 10 core tables hain: Complaints (55k+ records), ATMs (4k+ records), Alerts, Tasks, AuditLogs, Notifications, Messages, Reports, Documents, aur Users.